Cody / Planning Atlas

Screen journeys, a reusable character and the shared color system.

REVIEW FIRST.

PROTOTYPE NEXT.

A working atlas for the round display: first boot, essential OOBE, everyday use and future interactions.

Every screen has a writing area. Use it to review copy, touch targets, transitions and the character's response before implementation.



466 x 466

CURRENT FOCUS

Character rig, composable expressions and local UI. HTTP, PC integration and voice remain future design work.

A4 LANDSCAPE / ENGLISH / v1

38 entries + layout variants + expressions + 256 slots

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Screen inventory

Thirty-eight entries, with pause/end and sleep/wake expanded into separate review views.

01	Sleeping
02	Waking up
03	Ready
04	Hi, I'm Cody!
05	Let's pair
06	Scan to install
07	Pairing...
08	Try again
09	You're online.
10	Make it yours
11	Cody's color
12	Connect Codex
13	You're all set.
14	Looking for your PC...
15	Connected
16	Idle
17	Project active
18	Working...
19	Response ready
20	Paused / Session ended
21	PR pending
22	Checking...
23	Checks failed
24	Failed checks
25	Error details
26	Question
27	Choose an option
28	Recording...
29	Transcribing...
30	Review transcript
31	Reply sent
32	Connection lost
33	Low battery
34	Charging
35	Sleep / Wake
36	Settings
37	Rig view
38	Expression channels

Future connection, work, PR and voice screens are visual plans, not implemented integrations.

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The circle is usable space

Plan both the central square and the rounded regions.



CANVAS

466 x 466; center (233, 233); radius 233.

SAFE SQUARE

320 x 320 at (73, 73); bounds [73, 393).

BACK ICON

60 x 60 at (10, 203); center (40, 233).

ROUNDED REGIONS

73 px beside the square at its midline; about 9.38 px at its top and bottom.

ROTATION

Shared display/touch coordinates at 0/90/180/270 degrees. Back stays on the visual left.

NOTES

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Boot checkpoints

Three minimal frames make startup progress visible.

01 / Sleeping



Eyes closed. The first visible checkpoint starts once display, touch and saved orientation are ready.

NOTES

02 / Waking up



Eyes half open at a real intermediate startup boundary. This is not a measured 50% timer.

NOTES

03 / Ready



Eyes open when startup completes. This does not imply a network or PC connection.

NOTES

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Meet Cody and start pairing

A short introduction leads to one clear QR action.

04 / Hi, I'm Cody!



A minimal introduction. Cody fills the display before setup begins.

NOTES

05 / Let's pair



Explain the next QR step in one short message. Show QR is the primary action.

NOTES

06 / Scan to install



Install the pairing app, then continue. This sample QR uses an intentionally invalid example address.

NOTES

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Complete network setup

Waiting, retry and confirmation each have a clear purpose.

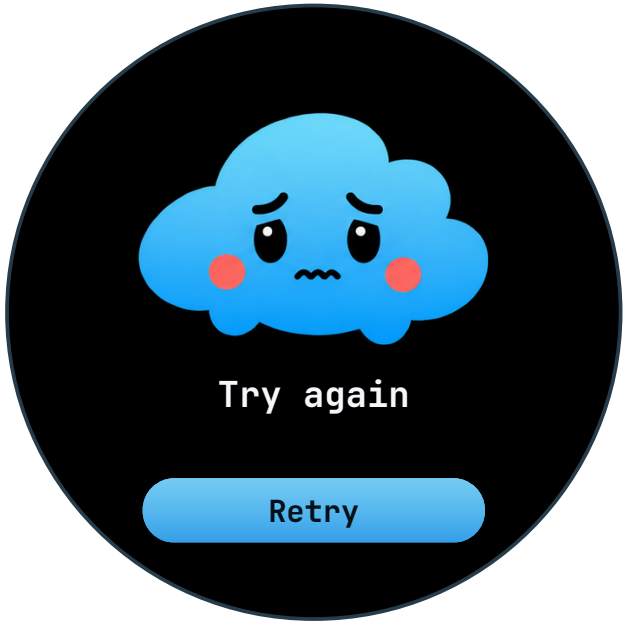
07 / Pairing...



Wait for setup on the phone. Keep cancellation available and show the real connection state.

NOTES

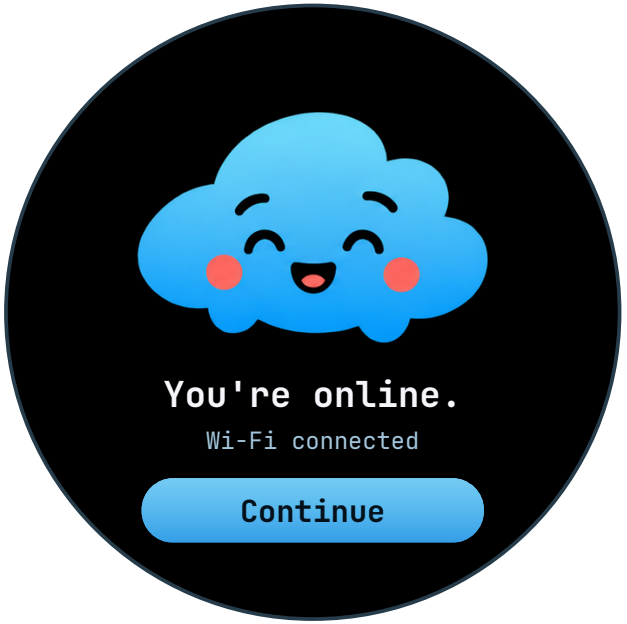
08 / Try again



A recoverable pairing failure. Retry returns to the relevant setup step.

NOTES

09 / You're online.



Confirm Wi-Fi association. The PC connection is a separate later milestone.

NOTES

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Preferences and PC setup

Make Cody yours, then plan the future PC connection.

10 / Make it yours



Wake, rotation and sound preferences. Manual 0/180 controls are shown in this compact setup view.

NOTES

11 / Cody's color



Choose Blue or Mint. The character engine shares one rig across both accent palettes.

NOTES

12 / Connect Codex



Finish setup on the PC. This atlas plans the screen; connection features are outside the current PoC.

NOTES

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Connect and return

First-run completion and later connection states.

13 / You're all set.



Show this confirmation only after the future application handshake succeeds.

NOTES

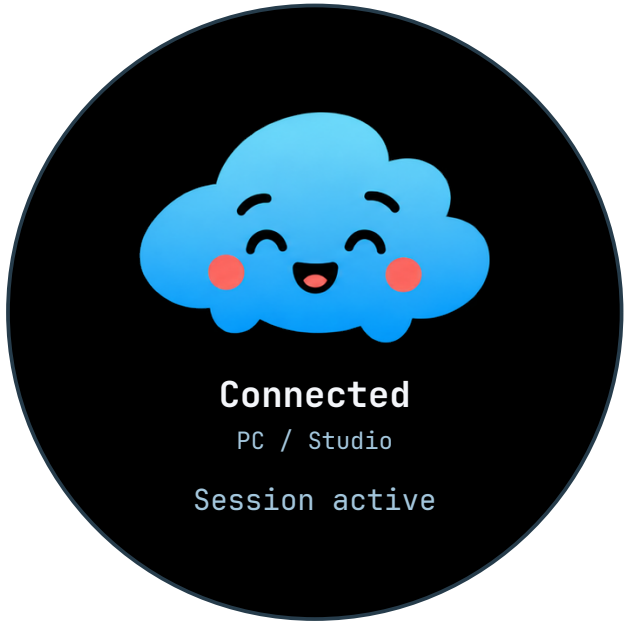
14 / Looking for your PC...



The network is available while the application connection is still pending.

NOTES

15 / Connected



A brief connection confirmation, followed by the active project or idle view.

NOTES

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Cody in everyday use

One reusable character, independent activity and expression channels.

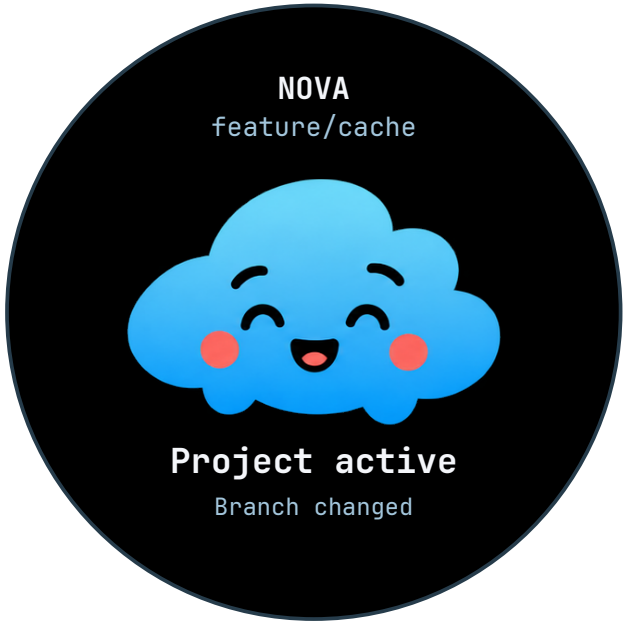
16 / Idle



Breathing, gaze and occasional blinks. Expressions combine independently instead of replacing the whole face.

NOTES

17 / Project active



Project, branch and accent identify the selected context. Branch changed is a variant of this layout.

NOTES

18 / Working...



The same layout supports Reading, Editing and Running. Activity does not force one exclusive expression.

NOTES

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Respond, pause and rest

Activity status remains distinct from success or connection health.

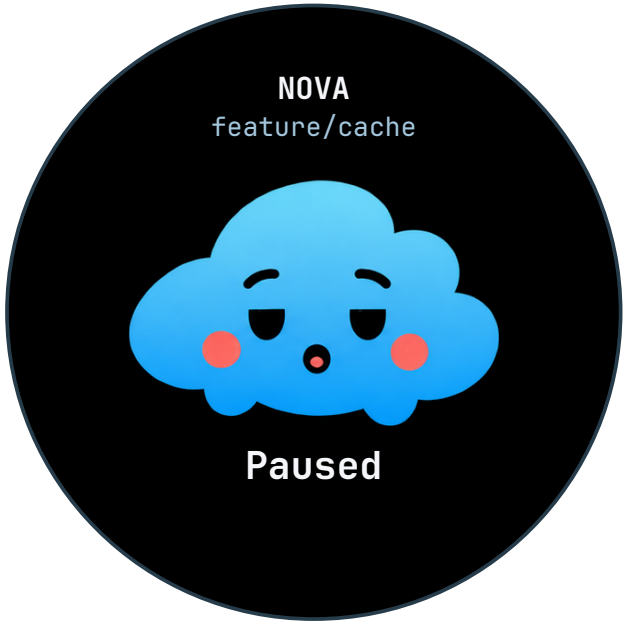
19 / Response ready



Invite the user to read the response. This event alone does not certify task or test success.

NOTES

20A / Paused



Work has paused. Cody relaxes while preserving the selected project and any pending interaction.

NOTES

20B / Session ended



The session is over. Cody returns to rest; connection and battery remain separate states.

NOTES

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Follow a pull request

Future result views use concise status and verified progress.

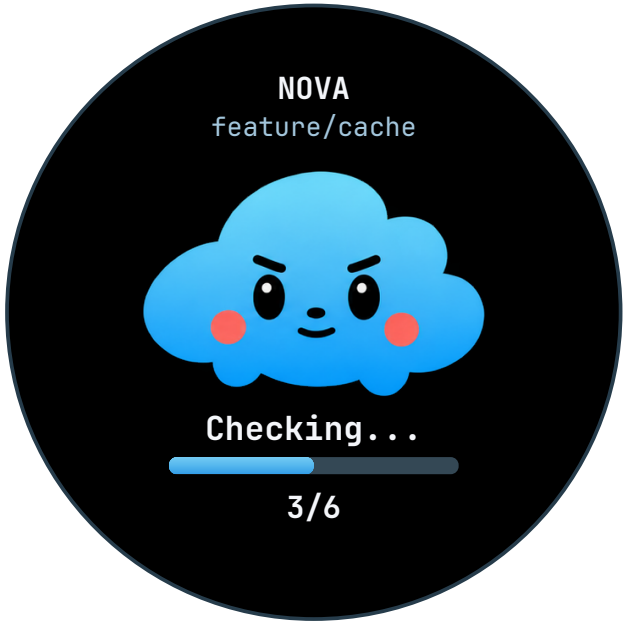
21 / PR pending



A pull request is waiting for results. All project and PR values in this atlas are illustrative.

NOTES

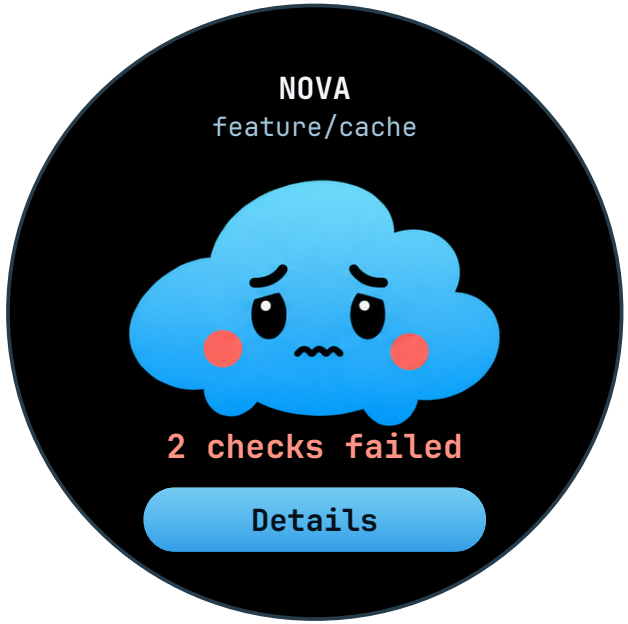
22 / Checking...



Show a count only when real progress is available. The sample here is 3 of 6 checks.

NOTES

23 / Checks failed



Summarize the failures and offer Details. A worried expression supports the explicit status text.

NOTES

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Read details and open a question

Content gets the central area when it needs to be read.

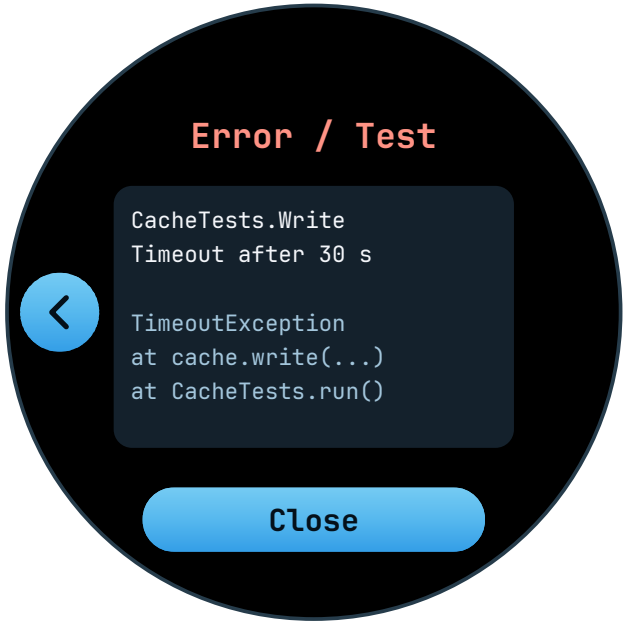
24 / Failed checks



A readable list of failures leads to the selected log. Content scrolls in one main region.

NOTES

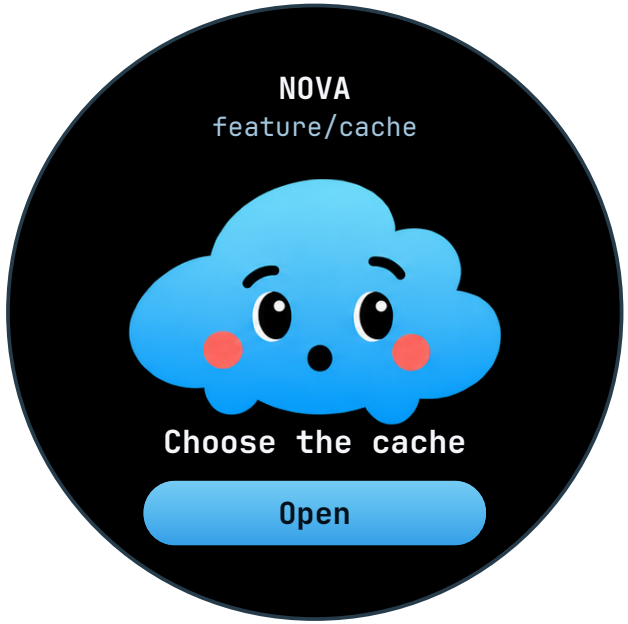
25 / Error details



Give the log the central area. A small mascot can yield all its space when reading needs it.

NOTES

26 / Question



A compact attention cue opens the full question. Later activity must not dismiss an unanswered question.

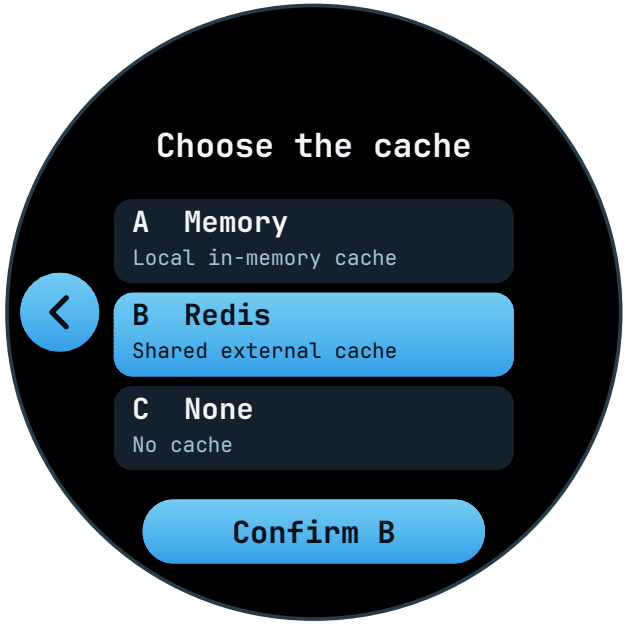
NOTES

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Choose or speak

Explicit choices and a future voice flow share the same interaction model.

27 / Choose an option



Select an answer, read its description, then confirm explicitly. No answer is chosen on timeout.

NOTES

28 / Recording...



Push to talk with elapsed duration. The waveform in this planning view is illustrative.

NOTES

29 / Transcribing...



Show a distinct waiting state after recording, with cancellation available.

NOTES

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Review, acknowledge and recover

A reply is deliberate; connection loss preserves pending context.

30 / Review transcript



Read before sending. Record again returns to capture and avoids requiring a tiny on-device keyboard.

NOTES

31 / Reply sent



A short acknowledgment after the future receiver confirms delivery, then return to context.

NOTES

32 / Connection lost



Keep pending content intact while reconnecting. Do not infer connection health from facial expression.

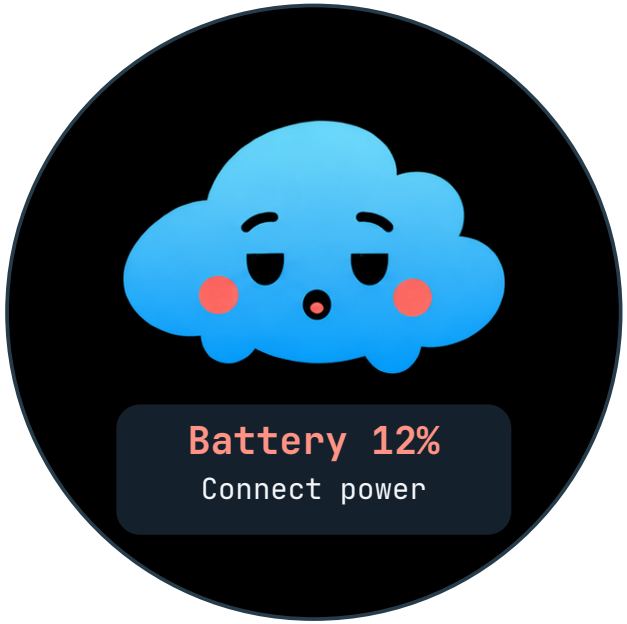
NOTES

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Battery and sleep

Device state changes the presentation without discarding useful content.

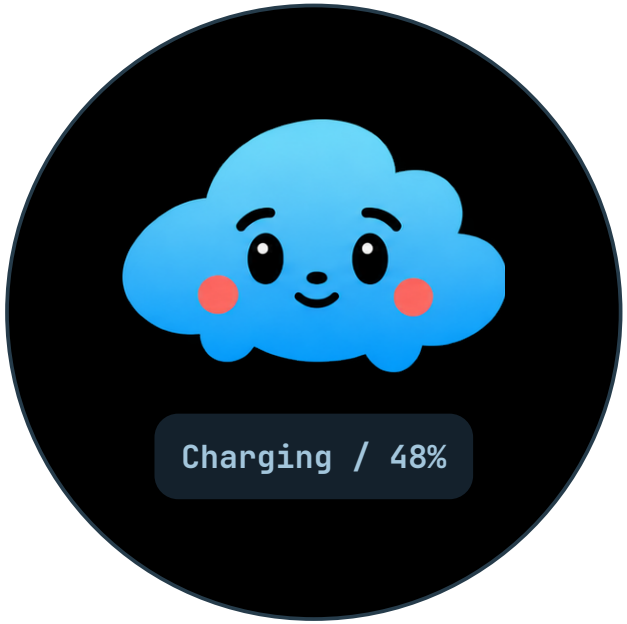
33 / Low battery



A real low-charge event adds a compact notice. On question screens it must preserve the question.

NOTES

34 / Charging



Show charging as an independent device indicator. Energy and facial channels can still vary.

NOTES

35A / Sleep



Eyes close while the mouth and cloud continue a slow animated breath, before any display-off policy.

NOTES

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Wake, settings and the rig

The first PoC focuses on local UI and the reusable character.

35B / Wake



Eyes reopen and attention returns. Restore the previous useful screen after the wake transition.

NOTES

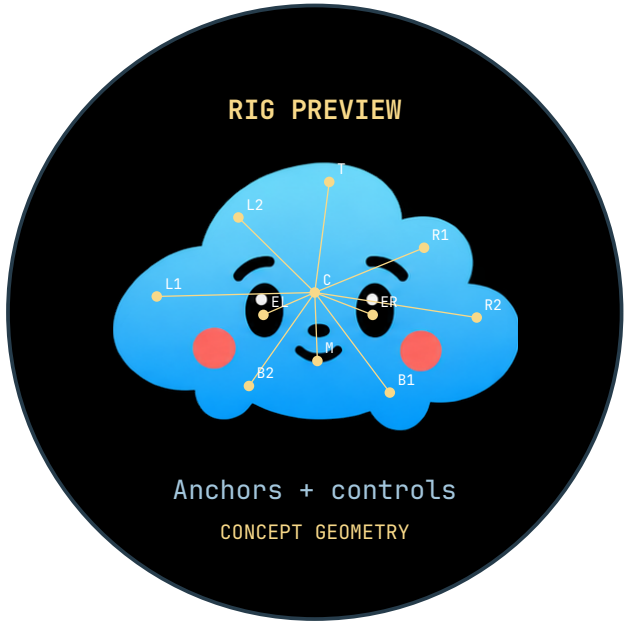
36 / Settings



Revisit OOBE preferences later. Brightness and additional controls can be reviewed during planning.

NOTES

37 / Rig view



Illustrative anchor overlay: cloud lobes, face center and facial anchors. Final topology is still to be designed.

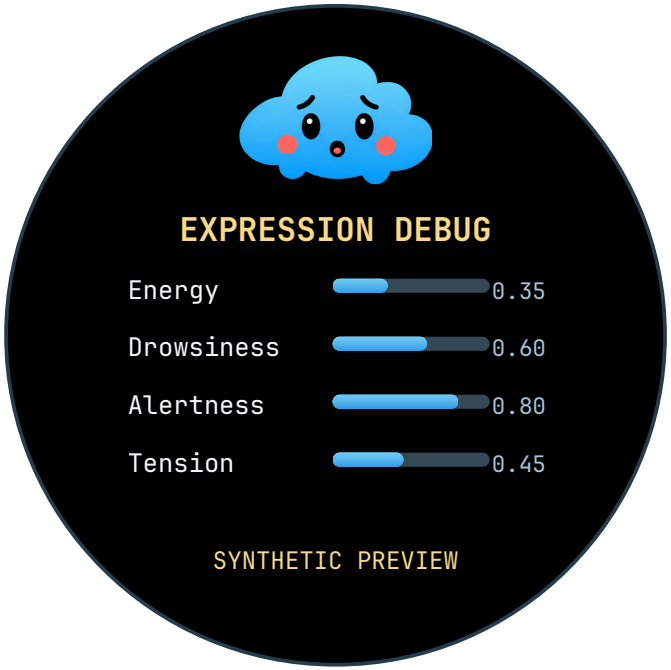
NOTES

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Inspect the composed expression

Debug shows the same evaluated pose that the character renderer uses.

38 / Expression channels



ONE CHARACTER, MANY CHANNELS

Inputs

Valid local events: motion, touch, orientation, charge and time since interaction.

Expressive state

Energy, drowsiness, alertness, tension and mood can coexist.

Independent motion

Breathing, gaze and blinking compose with the selected expression.

Shared geometry

Rig anchors and curve handles come from the same evaluated pose.

Preview controls

Synthetic examples are confined to a clearly labelled desktop preview.

NOTES

NOTES

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Expression combinations / 01

Illustrative moments made from independent channels; these are not exclusive face states.



Calm

Relaxed brows + open eyes + gentle smile.

Slow breath; occasional blink.

NOTES



Curious

Raised brow + shifted gaze + small open mouth.

Keep the body breathing independently.

NOTES



Happy

Smiling eyes + lifted brows + open smile.

A brief response can settle back into idle.

NOTES



Focused

Attentive eyes + gathered brows + quiet mouth.

Subtle breath keeps the pose alive.

NOTES

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Expression combinations / 02

Illustrative moments made from independent channels; these are not exclusive face states.



Afraid

Wide eyes + worried brows + breathing mouth.

Alertness rises quickly, then eases down.

NOTES



Stressed

Tense brows + tired eyes + a tight mouth.

Tension can coexist with low energy.

NOTES



Sleeping

Closed eyelids + relaxed brows + open mouth.

Slow breath continues with closed eyes.

NOTES



Drowsy

Half-closed eyes + relaxed brows + open mouth.

A surprise may briefly reopen the eyelids.

NOTES

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The shared palette / all 256 slots

Exact RGB888 design swatches and indices from palette256.json.

000 #000000	001 #FFFFFF	002 #F4F6FA	003 #04101A	004 #0B1118	005 #111A24	006 #182330	007 #203040	008 #2A3C4D	009 #35495B	010 #40576A	011 #4D6477	012 #5B7184	013 #6B8092	014 #7D91A2	015 #92A3B2
016 #A5B4C1	017 #B9C6D1	018 #CBD6DF	019 #DCE4EA	020 #75CBF3	021 #74CAF3	022 #73CAF3	023 #72C9F3	024 #71C9F2	025 #70C8F2	026 #6FC7F2	027 #6EC7F2	028 #6DC6F2	029 #6CC6F2	030 #6BC5F1	031 #6AC4F1
032 #69C4F1	033 #68C3F1	034 #67C3F1	035 #66C2F1	036 #66C2F1	037 #65C1F0	038 #64C0F0	039 #63C0F0	040 #62BFF0	041 #61BFF0	042 #60BEF0	043 #5FBDEF	044 #5EBDEF	045 #5DBCEF	046 #5CBCEF	047 #5BBBEF
048 #5ABAEF	049 #59BAEE	050 #58B9EE	051 #57B9EE	052 #56B8EE	053 #55B7EE	054 #54B6EE	055 #53B6ED	056 #52B5ED	057 #51B4ED	058 #50B3ED	059 #4FB2EC	060 #4DB2EC	061 #4CB1EC	062 #4BB0EC	063 #4AAFEC
064 #49AEEB	065 #48AEEB	066 #47ADEB	067 #46ACEB	068 #45ABEA	069 #44AAEA	070 #43A9EA	071 #42A9EA	072 #41A8E9	073 #40A7E9	074 #3FA6E9	075 #3EA5E9	076 #3CA5E9	077 #3BA4E8	078 #3AA3E8	079 #39A2E8
080 #38A1E8	081 #37A1E7	082 #36A0E7	083 #359FE7	084 #9EF0D8	085 #9CEFD7	086 #9AEFD6	087 #97EED6	088 #95EED5	089 #93EDD4	090 #91ECD3	091 #8EECD2	092 #8CEBD2	093 #8AEBD1	094 #88EAD0	095 #85E9CF
096 #83E9CE	097 #81E8CD	098 #7FE8CD	099 #7CE7CC	100 #7AE6CB	101 #78E6CA	102 #76E5C9	103 #74E5C9	104 #71E4C8	105 #6FE3C7	106 #6DE3C6	107 #6BE2C5	108 #68E2C5	109 #66E1C4	110 #64E1C3	111 #62E0C2
112 #5FDFC1	113 #5DDFC1	114 #5BDEC0	115 #59DEBF	116 #57DDBE	117 #54DCBD	118 #52DCBD	119 #50DBBC	120 #4EDBBB	121 #4BDABA	122 #49D9B9	123 #47D9B8	124 #45D8B8	125 #42D8B7	126 #40D7B6	127 #3ED6B5
128 #3CD6B4	129 #39D5B4	130 #37D5B3	131 #35D4B2	132 #FF9385	133 #FE9183	134 #FE8E81	135 #FD8C7F	136 #FC8A7D	137 #FC877B	138 #FB8579	139 #FA8377	140 #FA8075	141 #F97E73	142 #F87C71	143 #F8796F
144 #F7776D	145 #F7746B	146 #F67269	147 #F57067	148 #F56D65	149 #F46B63	150 #F36961	151 #F3665F	152 #F2645D	153 #F1625B	154 #F15F59	155 #F05D57	156 #F8D989	157 #F6D685	158 #F5D281	159 #F3CF7D
160 #F1CC79	161 #F0C875	162 #EEC571	163 #ECC26D	164 #EBBE68	165 #E9BB64	166 #E7B860	167 #E6B45C	168 #E4B158	169 #E2AE54	170 #E1AA50	171 #DFA74C	172 R #000000	173 R #000000	174 R #000000	175 R #000000
176 R #000000	177 R #000000	178 R #000000	179 R #000000	180 R #000000	181 R #000000	182 R #000000	183 R #000000	184 R #000000	185 R #000000	186 R #000000	187 R #000000	188 R #000000	189 R #000000	190 R #000000	191 R #000000
192 R #000000	193 R #000000	194 R #000000	195 R #000000	196 R #000000	197 R #000000	198 R #000000	199 R #000000	200 R #000000	201 R #000000	202 R #000000	203 R #000000	204 R #000000	205 R #000000	206 R #000000	207 R #000000
208 R #000000	209 R #000000	210 R #000000	211 R #000000	212 R #000000	213 R #000000	214 R #000000	215 R #000000	216 R #000000	217 R #000000	218 R #000000	219 R #000000	220 R #000000	221 R #000000	222 R #000000	223 R #000000
224 R #000000	225 R #000000	226 R #000000	227 R #000000	228 R #000000	229 R #000000	230 R #000000	231 R #000000	232 R #000000	233 R #000000	234 R #000000	235 R #000000	236 R #000000	237 R #000000	238 R #000000	239 R #000000
240 R #000000	241 R #000000	242 R #000000	243 R #000000	244 R #000000	245 R #000000	246 R #000000	247 R #000000	248 R #000000	249 R #000000	250 R #000000	251 R #000000	252 R #000000	253 R #000000	254 R #000000	255 R #000000

172 allocated slots / 84 reserved black slots / R = reserved, not an extra black role

Design colors are sRGB. Several entries may map to the same RGB565 word; 256 slots do not imply 256 unique colors.

NOTES

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Color roles and soft gradients

The cloud and filled controls share the blue ramp.

00-03	**Fixed**		4 slots	Black, highlights, near-white text, near-black ink
04-19	**Neutrals**		16 slots	Surfaces, borders and secondary text
20-83	**Blue**		64 slots	Default Cody and button gradient
84-131	**Mint**		48 slots	Personalization allocation
132-155	**Coral**		24 slots	Cheeks and error emphasis
156-171	**Gold**		16 slots	Optional future status allocation
172-255	**Reserved**		84 slots	Unallocated; stored as black

BLUE ANCHORS `#75CBF3 -> #56B8EE -> #359FE7`

The RGB565 table occupies 512 bytes. Artwork studies are full-color references; runtime quantization remains future work.

NOTES

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Decisions before the PoC

Use this page to capture choices after reviewing the atlas.

- [ ] Cloud silhouette, anchor count and stable point names
- [ ] Eye, brow, mouth and breathing channels to prototype first
- [ ] Expression limits and event-to-expression mappings
- [ ] Screen copy, button sizes and use of rounded regions
- [ ] Palette allocation, font sizes and note-worthy color changes

Sources: approved OOBE v5, original journey references, repository layout rule and palette256.json.

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Separate app installation from pairing

A01 supplements the printed atlas. Existing pages and screen IDs stay in place.

REQUESTED BY	AFFECTED BASE PAGES	PRINT ONLY THIS PAGE
Umberto Giacobbi (project owner)	**05 / 06 / 21**	**23 / keep the existing printout**

06 / REVISED



Installation QR. Remove the visible 'Scan to install' title. Pair Cody opens the device QR.

06-bis / NEW



Device QR. Scan it inside the provisioning app. Back stops the active BLE session and returns to 06.

WHY THIS CHANGE

The installation link downloads the app. A second QR identifies Cody for Wi-Fi provisioning over BLE. One QR cannot express both steps clearly.

UPDATED JOURNEY

05 Show QR -> 06
06 Pair Cody -> 06-bis
Credentials received -> 07
Failure -> 08 / Retry -> 06-bis
Wi-Fi connected + IP -> 09

SHARED ACTION BUTTON

Reuse one capsule control: Blue / Primary, Mint / Success, Gold / Warning, Coral / Danger, Neutrals / Secondary. Use the palette roles on base page 21; Retry stays Primary.

QR drawings are illustrative. Firmware must supply the real app URL and device-session payload.

06-bis is a screen ID, not a PDF page number. This addendum records the owner's requested design change.

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A02 / Screen 11 becomes Cody's look

One scrollable single-choice list changes the live preview above it.



SCREEN 11 / REVISED

Affected historical pages: 02 inventory and 07 OOBE layouts. The old Blue/Mint color chips are superseded; their printed pages remain historical.

Five authored looks

Pearl Mist DEFAULT
`cody.skin.pearl_mist.v1`

Aurora Veil
`cody.skin.aurora_veil.v1`

Cosmic Stardust
`cody.skin.cosmic_stardust.v1`

Enchanted Mosaic
`cody.skin.enchanted_mosaic.v1`

Dreamwave
`cody.skin.dreamwave.v1`

Interaction contract

Drag vertically to scroll. Tap one row to select exactly one look. The preview changes immediately. The user sees the cool name; firmware stores only the stable internal ID. Back or Done returns to the caller. No color editor, mixing or texture controls are added.

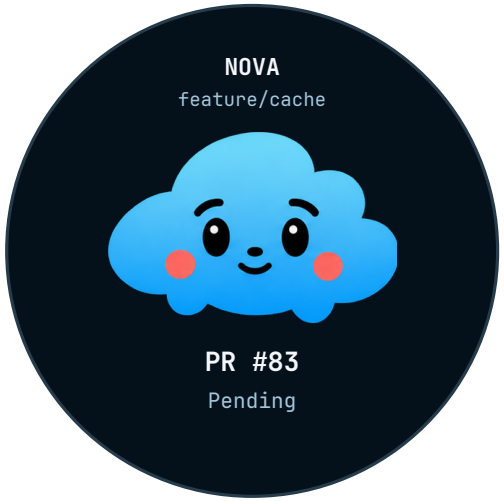
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A03 / Pull requests become status-only

Screens 21-24 stay concise and read-only; screen 25 is retired.

PRINT PAGE 25 ONLY

21 / PR pending



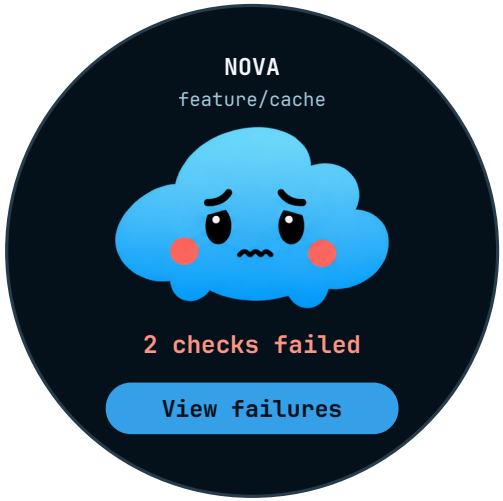
Observe only. No PR action is offered.

22 / Checks running



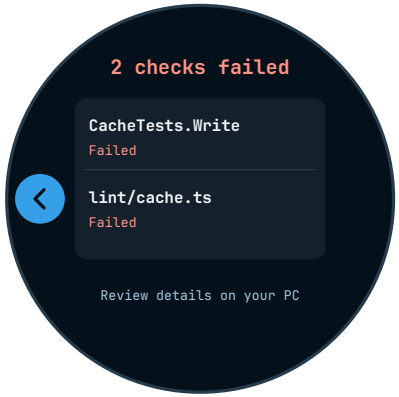
Show a count only when the exact total is known.

23 / Checks failed



One explicit action opens the failure list.

24 / Failed checks



READ-ONLY CONTRACT

Screen 22 briefly changes to 'All checks passed' before returning to screen 17. Screen 24 lists real failed check names and uses the fixed lateral Back control. It has no View log button and no selectable log rows.

25 / RETIRED — Raw logs and stack traces stay on the PC. Cody does not merge, approve, close, rerun or comment on a PR.

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A04 / Pairing handoff and continued setup

One temporary Debug screen; all other steps keep their existing atlas IDs.

PRINT PAGE 26 ONLY

39 / Pairing debug

DEBUG ONLY



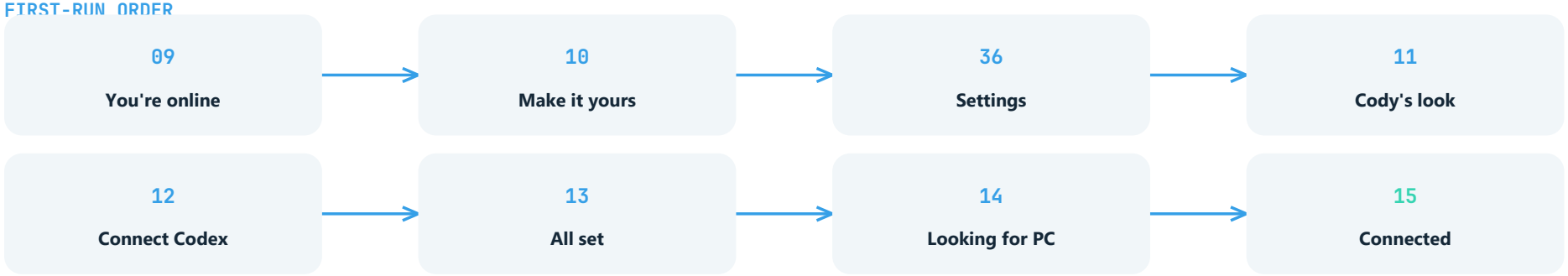
PAIRING SUCCESS CONTRACT

A healthy current Wi-Fi/IP handoff writes cody_firstboot=false and reads it back before any success UI. Debug builds then show screen 39. One awake tap dismisses it. Release builds contain no screen-39 route and go directly to 09.

SCREEN 11 / REVISED INTERACTION

Use the five-look selector shown on planning page 24. Remove Back. Pearl Mist remains visibly selected by default, but Done starts disabled. The user must tap one row explicitly; tapping Pearl Mist itself is a valid confirmation. The selector remains reusable from screen 36 later.

FIRST-RUN ORDER



CONFIG_LUNA_SHELL_CODY_ENABLE_CODEX=y includes 12; n skips from 11 to 13.

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